[T]-Theory: Complex Systems

P3 · Formal Foundation · For: The Network Researcher

Scale-Invariant Propagator — the same fixed point across 61 orders of magnitude.

The RG fixed point:

- USF is scale-invariant under RG flow: $G \rightarrow G$ (proved)
- $k(\sigma) = k_0/\lambda^\sigma$ — geometric coupling decay
- Wave equation satisfied at every scale level (physlib, proved)
- The theory describes its own propagation (P23 fixed-point)

Network correspondence:

- Network adjacency $A_{ij} \leftrightarrow$ coupling matrix W_8
- Percolation threshold $\leftrightarrow$ consciousness threshold T_c
- Giant component $\leftrightarrow$ global field integration
- Cascade dynamics $\leftrightarrow$ somatic contagion waves

I · The Scale-Invariant Propagator

The RG flow of the USF coupling:

$$k(\sigma) = \frac{k_0}{\lambda^\sigma}, \quad \lambda > 1, \quad \sigma \in \{0, \dots, 19\}$$

The key invariance: the wave equation $(\nabla^2 + k^2)G = \delta$ is satisfied at every scale level simultaneously. Proved via `geometricFlow_waveEquation`.

The **Zoom Operator** Z_σ maps scale σ to $\sigma + 1$ preserving the field equation. Scale invariance is *derived*, not assumed.

II · Power Laws and Fixed Points

Near the consciousness threshold T_c (scale 7):

$$\xi = \frac{1}{k} \rightarrow \infty, \quad C(r) \sim r^{-(d-2+\eta)}$$

This is standard critical phenomena language. The USF gives: $\eta = 0$ at leading order (mean-field). Corrections from X_7 moduli geometry.

The **fixed-point identity** (P23): the USF propagator G is itself an instance of a field equation. The theory is its own attractor.

III · Complex Systems as Field Theory

Emergent complexity arises when the field exits the perturbative regime:

- **Phase transition**: $k \rightarrow 0$, long-range correlations onset
- **Synchronisation**: frequency locking as attractor basin overlap
- **Cascade failure**: avalanche = subcritical field discharge
- **Self-organised criticality**: USF operates at scale-7 critical point

IV · The 20-Scale Network Hierarchy

Each scale is a network embedded in the next: neurons embed in brains, brains in organisms, organisms in swarms, swarms in societies. The USF propagator is the inter-network kernel at each embedding.

$G_{\sigma\rightarrow\sigma+1}$ = the propagator from scale σ nodes to scale $\sigma + 1$ effective degrees of freedom.

G-ID: *Scale-Invariant Propagator* — G invariant under renormalisation group flow. ORCID: 0009-0007-2194-0850 · · Zurich 2026